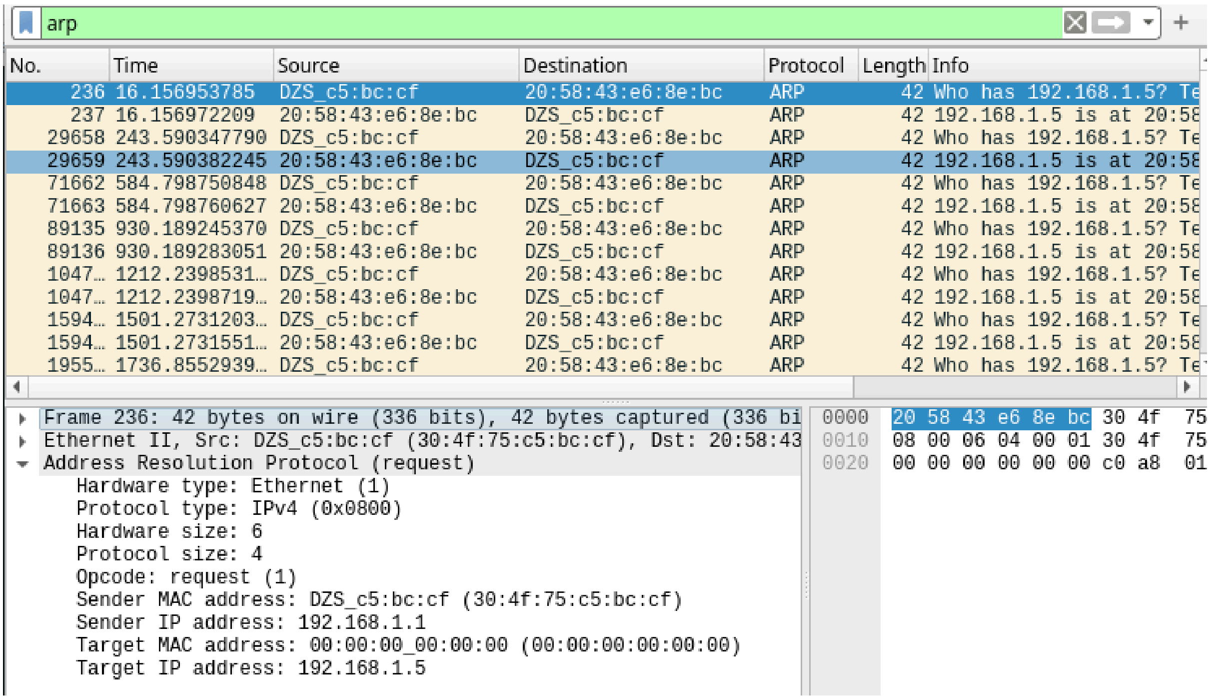
Manish Raj M S S
Thiagarajar College Of Engineering

Networking Assignment
Week 4

-Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address.



arp

No.	Time	Source	Destination	Protocol	Length	Info
236	16.156953785	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
237	16.156972209	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
29658	243.590347790	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
29659	243.590382245	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
71662	584.798750848	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
71663	584.798760627	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
89135	930.189245370	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
89136	930.189283051	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
1047…	1212.2398531…	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
1047…	1212.2398719…	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
1594…	1501.2731203…	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te
1594…	1501.2731551…	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	192.168.1.5 is at 20:58
1955…	1736.8552939…	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	Who has 192.168.1.5? Te

```
Frame 236: 42 bytes on wire (336 bits), 42 bytes captured (336 bi
Ethernet II, Src: DZS_c5:bc:cf (30:4f:75:c5:bc:cf), Dst: 20:58:43
Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: DZS_c5:bc:cf (30:4f:75:c5:bc:cf)
    Sender IP address: 192.168.1.1
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.5
```

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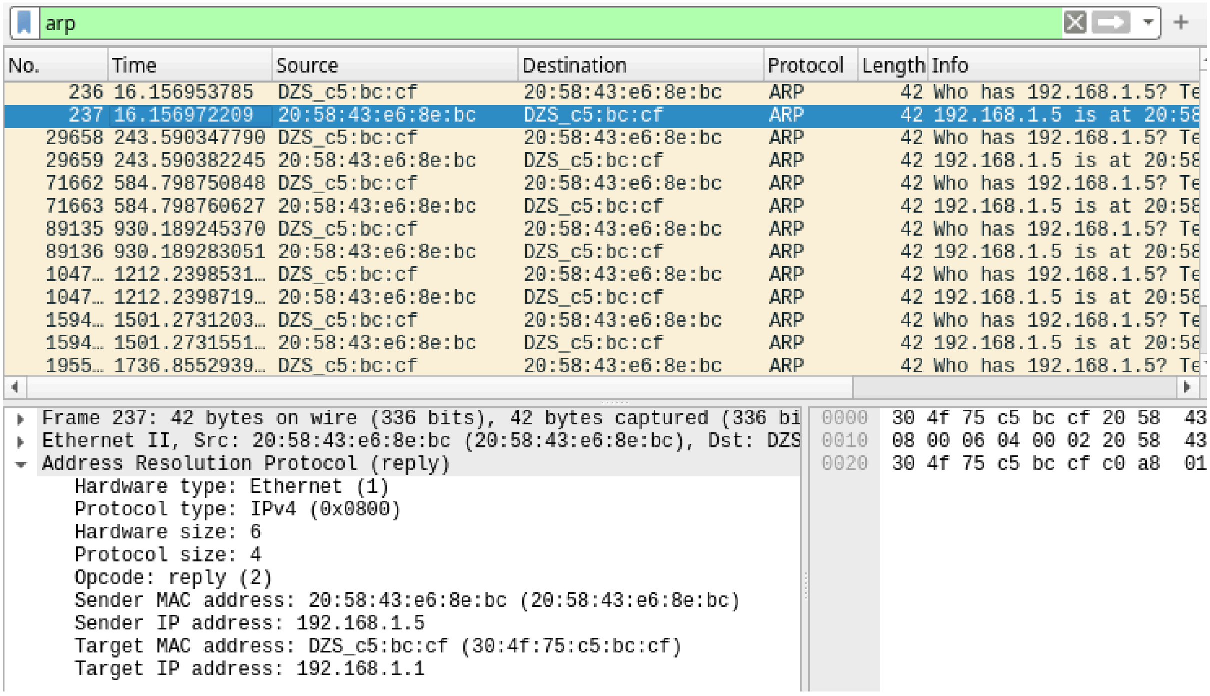
0000  20 58 43 e6 8e bc 30 4f  75
0010  08 00 06 04 00 01 30 4f  75
0020  00 00 00 00 00 00 c0 a8  01
```



arp

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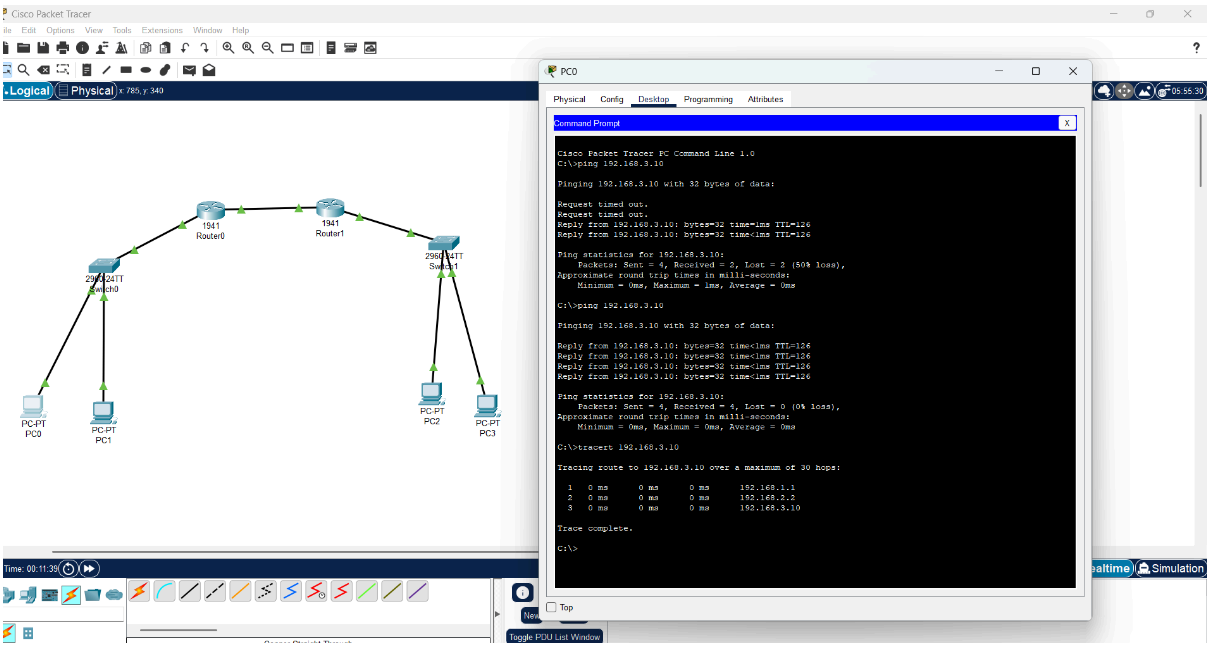
```
Frame 237: 42 bytes on wire (336 bits), 42 bytes captured (336 bi
Ethernet II, Src: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc), Dst: DZS
Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc)
    Sender IP address: 192.168.1.5
    Target MAC address: DZS_c5:bc:cf (30:4f:75:c5:bc:cf)
    Target IP address: 192.168.1.1
```

```
0000  30 4f 75 c5 bc cf 20 58  43
0010  08 00 06 04 00 02 20 58  43
0020  30 4f 75 c5 bc cf c0 a8  01
```

- ARP packets were captured and analyzed using Wireshark.
- When the device attempted to communicate with the router, it generated an ARP request.
- The ARP request is a broadcast frame sent to all devices in the network, asking “Who has 192.168.1.1?”.
- The request contains the sender’s IP and MAC address, while the target MAC address is unknown (set to 00:00:00:00:00:00).
- The router responded with an ARP reply, which is a unicast frame sent directly to the requesting device.
- The reply contains the router’s MAC address corresponding to its IP address (192.168.1.1).
- This allows the device to update its ARP table and use the MAC address for further communication.
- ARP plays an important role in packet forwarding by mapping IP addresses to MAC addresses.
- Since communication within a local network happens using MAC addresses, ARP ensures that packets are delivered to the correct device.
- Without ARP, devices would not be able to identify the destination hardware address, and communication within the network would fail.

-Manually configure static routes on a router to direct packets to different subnets. Use the ip route command and verify connectivity using ping and traceroute.



- Static routes were manually configured on the routers using the ip route command.
- The network consisted of multiple subnets connected through routers.
- Each router was configured with static routes to reach remote networks.
- Connectivity was verified using the ping command between devices in different subnets.

- Initially, some packets were lost due to ARP resolution, but subsequent pings were successful.
- Traceroute was used to observe the path taken by packets.
- The output showed that packets passed through intermediate routers before reaching the destination.
- This confirms that static routing was correctly configured and functioning properly.

-Given a network address of 10.0.0.0/24, divide it into 4 equal subnets.

Calculate the new subnet mask.
Determine the valid host range for each subnet.
Assign IP addresses to devices in Packet Tracer and verify connectivity.

Given Network: 10.0.0.0/24 (Subnet Mask: 255.255.255.0)

To divide the network into 4 equal subnets, 2 bits are borrowed from the host portion.

New Subnet Mask: 255.255.255.192 (/26)

Each subnet contains 64 IP addresses, out of which 62 are usable for hosts.

Subnet 1:
Network Address: 10.0.0.0/26
Valid Host Range: 10.0.0.1 – 10.0.0.62
Broadcast Address: 10.0.0.63

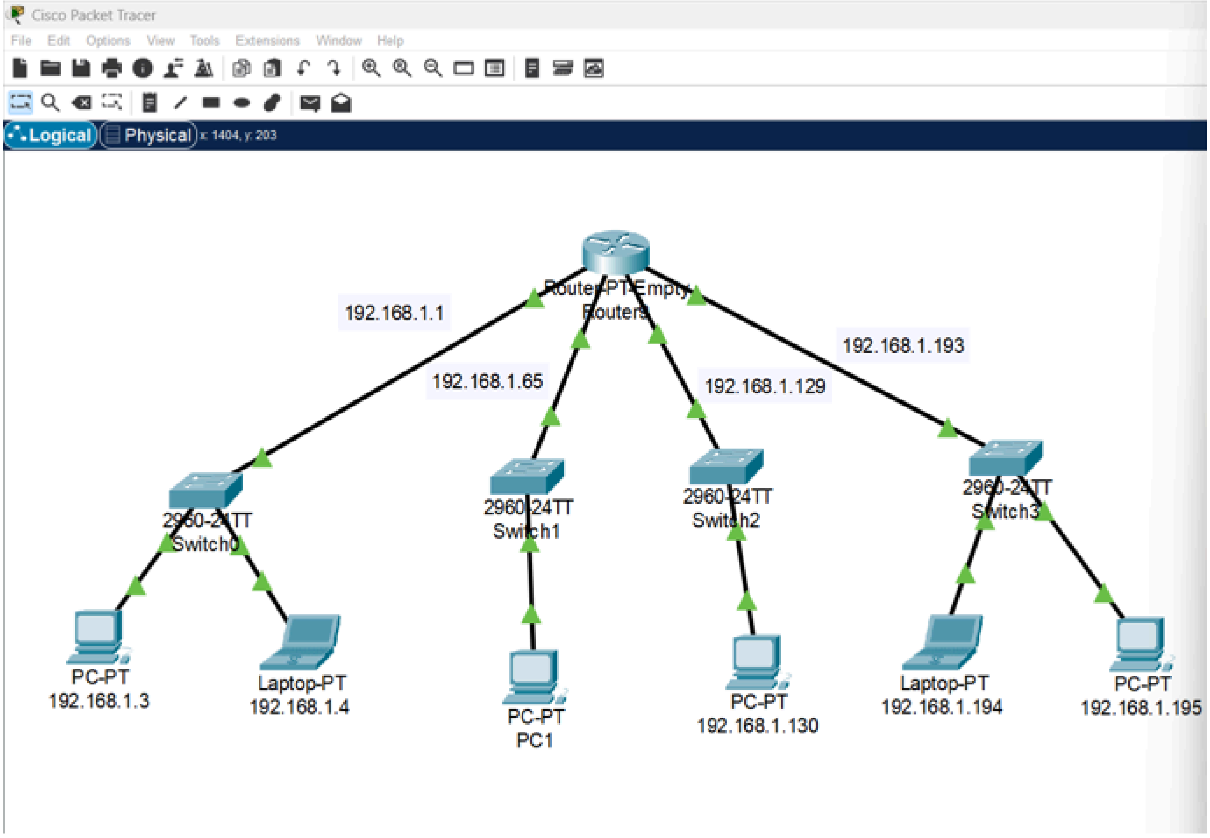
Subnet 2:
Network Address: 10.0.0.64/26
Valid Host Range: 10.0.0.65 – 10.0.0.126
Broadcast Address: 10.0.0.127

Subnet 3:
Network Address: 10.0.0.128/26
Valid Host Range: 10.0.0.129 – 10.0.0.190
Broadcast Address: 10.0.0.191

Subnet 4:
Network Address: 10.0.0.192/26
Valid Host Range: 10.0.0.193 – 10.0.0.254
Broadcast Address: 10.0.0.255

IP addresses were assigned to devices in Packet Tracer from these subnets.

Connectivity was verified using ping, and devices within the same subnet were able to communicate successfully.



-You are given three IP addresses: 192.168.10.5, 172.20.15.1, and 8.8.8.8. Identify the class of each IP address. Determine if it is private or public. Explain how NAT would handle a private IP when accessing the internet.

1) 192.168.10.5
Class: C
Default Subnet Mask: 255.255.255.0
Type: Private IP

2) 172.20.15.1
Class: B
Default Subnet Mask: 255.255.0.0
Type: Private IP

3) 8.8.8.8
Class: A
Default Subnet Mask: 255.0.0.0
Type: Public IP

NAT Explanation:

Private IP addresses cannot directly access the internet.

When a device with a private IP sends data to the internet, Network Address Translation (NAT) is used.

Before NAT, the source IP is the private IP address (e.g., 192.168.x.x or 172.x.x.x).

The router translates this private IP into its public IP address.

After NAT, the source IP becomes the public IP of the router.

This allows multiple devices in a private network to communicate with external networks using a single public IP address.